F70LA Life Insurance Mathematics A (2025/26)
Tutorial 8

1. Today, a man aged 45 buys a 15-year endowment insurance with a basic sum assured of £100,000. At the end of each policy year, while the insured is alive, the sum assured increases at the simple rate of 6% per annum. In other words, the sum assured increases at the end of each policy year by £6,000. Death benefits are payable at the end of year of death.

   (a) State the death benefit if the man dies during the first policy year.

   (b) Calculate the death benefit if the man dies during the second policy year.

   (c) Calculate the death benefit if the man dies during the last policy year.

   (d) Calculate the survival benefit if the man survives to age 60.

   (e) Calculate the expected present value of the endowment insurance using the basis AM92 Ultimate at 4% per annum effective (see the last pages of this tutorial).

2. Today, a man aged 45 buys a 20-year endowment insurance with a basic sum assured of £100,000. At the end of each policy year, while the insured is alive, the sum assured (i.e. the basic sum assured plus any earlier increases) increases by 1.923% per annum. Death benefits are payable at the end of year of death.

   (a) State the death benefit if the man dies during the first policy year.

   (b) Write down an expression for the death benefit involving a power of 1.01923, if the man dies during the second policy year.

   (c) Write down an expression for the death benefit involving a power of 1.01923, if the man dies during the last policy year.

   (d) Write down an expression for the survival benefit involving a power of 1.01923, if the man survives to age 65.

   (e) Calculate the expected present value of the endowment insurance using the basis AM92 Ultimate at 6% per annum effective.

3. A survival model is given by ELT15 Females (see the last pages of this tutorial). Assume an effective annual rate of interest of 4% for the calculations below.

   (a) Verify by direct calculation that $\ddot{a}_{80:\overline{5}|} = 4.0829$.

   (b) Using Woolhouse's formula, calculate $\ddot{a}^{(12)}_{80:\overline{5}|}$.

4. On the basis of the AM92 survival model with 4% per year interest (see the last pages of this tutorial) and using Woolhouse's formula. , calculate the values of

$$\text{(a) } \ddot{a}^{(4)}_{70}; \quad \text{(b) } \ddot{a}^{(4)}_{80}, \quad \text{(c) } \ddot{a}^{(4)}_{70:\overline{10}|}; \quad \text{(d) } \bar{a}_{70:\overline{10}|} \quad \text{and} \quad \text{(e) } \bar{A}_{70:\overline{10}|}.$$

5. Using Woolhouse's formula, calculate the values of (a) $\ddot{a}_{60:\overline{30|}}$; and (b) $A_{60:\overline{30|}}$, at an effective interest rate of 4% per year, given

$$\bar{a}_{60} = 10.628, \qquad \bar{a}_{90} = 2.689 \qquad \text{and} \qquad {}_{30}p_{60} = 0.03861.$$

6. (a) Show that, using Woolhouse's formula,

$$\ddot{a}^{(m)}_{x:\overline{1|}} \approx \frac{m+1}{2m} + \frac{m-1}{2m}\, v\, p_x.$$

   (b) An annuity is payable m-thly in advance throughout the lifetime of (x). The m-thly payments are constant throughout each year but increase by a factor $(1+r)$ at the start of each year. In the first year, the m-thly payments are each of amount $1/m$, so that the rate of payment is 1 unit per year. In the second year, each m-thly payment is of amount $(1+r)/m$, in the third year $(1+r)^2/m$, and so on, so that in the nth year, each m-thly payment is of amount $(1+r)^{n-1}/m$ for $n = 1, 2, 3, \ldots$.

      (i) Show that, valued at a constant effective rate of interest i per year, the expected present value of the annuity is

$$\sum_{n=0}^{\infty} (1+r)^n\, v^n\, {}_np_x\, \ddot{a}^{(m)}_{x+n:\overline{1|}}.$$

      (ii) Using part (a) above, or otherwise, show that this expression is approximately equal to

$$\frac{m+1}{2m}\, \ddot{a}^{\star}_x + \frac{m-1}{2m}\, \frac{\ddot{a}^{\star}_x - 1}{1+r},$$

      in which $\ddot{a}^{\star}_x$ equals $\ddot{a}_x$ evaluated at the effective rate of interest $\frac{i-r}{1+r}$.

7. A man aged 65 buys a guaranteed annuity policy, with a 5-year guarantee, to provide an income in retirement. This is a life annuity contract in which the payments for the first 5 years are guaranteed to be paid, regardless of whether the man is alive or not during that time. If the man survives for at least 5 years, the annuity payments continue until he dies. The annuity payments of £12,000 per annum are paid monthly in advance.

   The policy is paid by a single premium £P.

   The premium basis is as follows.

Interest:	6% per annum effective.
Mortality:	AM92 Select.
Expenses:	5% of the premium, and
	£150 at the time of each annuity payment.

   (a) Calculate the EPV of the annuity payments.

   (b) Write down an expression for the EPV of the expenses as a function of P.

(c) By equating the premium (i.e. the money coming into the insurance company) with the EPV of the benefits and expenses (i.e. the money paid out by the insurance company), calculate the single premium for the contract.

8. A man aged 45 buys a deferred annuity policy. The policy has premiums £P per annum payable annually during his lifetime for at most twenty years. If he survives to age 65, an annuity of £4 000 per year will be payable annually in advance from that age. If he dies before age 65, all premiums paid will be returned at the end of the year of death; this is a return-of-premium benefit.

   The premium basis is as follows.

Interest:	4% per annum effective.
Mortality:	AM92 Ultimate.
Expenses:	2% of each annual premium and £10 at the time of each annuity payment.

   (a) Calculate the EPV of the annuity payments.

   (b) Calculate $(IA)^{1}_{45:\overline{20}|}$.

   (c) Write down an expression for the EPV of the return-of-premium benefit as a function of P.

   (d) Write down an expression for the EPV of the expenses as a function of P.

   (e) By equating the EPV of the premiums (i.e. the money coming into the insurance company) with the EPV of the benefits and expenses (i.e. the money paid out by the insurance company), calculate the annual premium.

Numerical solutions: Q1(a) £100,000; Q1(b) £106,000; Q1(c) £184,000; Q1(d) £190,000; Q1(e) £105,350.50. Q2(a) £100,000; Q2(b) £$100,000 \times 1.0923^{1}$; Q2(c) £$100,000 \times 1.0923^{19}$; Q2(d) £$100,000 \times 1.0923^{20}$; Q2(e) £46,886.45. Q3(b) 3.8826.Q4(a) 10.000; Q4(b) 6.443; Q4(c) 7.154; Q4(d) 7.084; Q4(e) 0.7222; Q5(a) 11.090; Q5(b) 0.5735. Q7(a) £123,255.20; Q7(b) $0.05P + 18,488.28$; Q7(c) £149,203.66. Q8(a) £20,169.57; Q8(b) 0.75724; Q8(c) £$0.75724P$; Q8(d) £$(0.2756P + 50.42387)$; Q8(e) £1,586.24.

ELT15 (Females)

x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
0	100 000	632	0.006 32		78.956	0
1	99 368	55	0.000 55	0.000 73	78.462	1
2	99 313	30	0.000 30	0.000 35	77.505	2
3	99 283	22	0.000 22	0.000 25	76.528	3
4	99 261	18	0.000 18	0.000 20	75.545	4
5	99 243	15	0.000 16	0.000 17	74.559	5
6	99 228	15	0.000 15	0.000 15	73.570	6
7	99 213	14	0.000 14	0.000 14	72.581	7
8	99 199	14	0.000 14	0.000 14	71.591	8
9	99 185	13	0.000 13	0.000 14	70.601	9
10	99 172	13	0.000 13	0.000 13	69.610	10
11	99 159	14	0.000 14	0.000 14	68.620	11
12	99 145	14	0.000 14	0.000 14	67.629	12
13	99 131	15	0.000 15	0.000 14	66.638	13
14	99 116	18	0.000 18	0.000 17	65.649	14
15	99 098	21	0.000 22	0.000 20	64.660	15
16	99 077	26	0.000 26	0.000 24	63.674	16
17	99 051	31	0.000 31	0.000 29	62.691	17
18	99 020	31	0.000 31	0.000 31	61.710	18
19	98 989	32	0.000 32	0.000 32	60.729	19
20	98 957	31	0.000 31	0.000 32	59.748	20
21	98 926	32	0.000 32	0.000 32	58.767	21
22	98 894	32	0.000 33	0.000 32	57.786	22
23	98 862	33	0.000 33	0.000 33	56.805	23
24	98 829	32	0.000 33	0.000 33	55.823	24
25	98 797	34	0.000 34	0.000 33	54.842	25
26	98 763	34	0.000 35	0.000 34	53.860	26
27	98 729	35	0.000 36	0.000 35	52.878	27
28	98 694	38	0.000 38	0.000 37	51.897	28
29	98 656	39	0.000 40	0.000 39	50.917	29
30	98 617	43	0.000 43	0.000 42	49.937	30
31	98 574	46	0.000 47	0.000 45	48.958	31
32	98 528	51	0.000 52	0.000 50	47.981	32
33	98 477	57	0.000 57	0.000 54	47.006	33
34	98 420	61	0.000 63	0.000 60	46.032	34
35	98 359	68	0.000 69	0.000 66	45.061	35
36	98 291	74	0.000 75	0.000 72	44.092	36
37	98 217	81	0.000 82	0.000 79	43.124	37
38	98 136	88	0.000 90	0.000 86	42.160	38
39	98 048	96	0.000 98	0.000 94	41.197	39
40	97 952	105	0.001 07	0.001 02	40.237	40
41	97 847	114	0.001 17	0.001 12	39.279	41
42	97 733	126	0.001 29	0.001 23	38.325	42
43	97 607	138	0.001 42	0.001 35	37.374	43
44	97 469	154	0.001 58	0.001 49	36.426	44
45	97 315	173	0.001 77	0.001 67	35.483	45
46	97 142	192	0.001 98	0.001 87	34.545	46
47	96 950	212	0.002 19	0.002 08	33.612	47
48	96 738	234	0.002 41	0.002 30	32.685	48
49	96 504	257	0.002 66	0.002 53	31.763	49
50	96 247	283	0.002 94	0.002 80	30.846	50
51	95 964	312	0.003 26	0.003 10	29.936	51
52	95 652	342	0.003 57	0.003 42	29.032	52
53	95 310	372	0.003 90	0.003 74	28.134	53
54	94 938	406	0.004 28	0.004 08	27.242	54

ELT15 (Females)

x	l_x	d_x	q_x	μ_x	$\mathring{e}_x$	x
55	94 532	450	0.004 75	0.004 51	26.357	55
56	94 082	499	0.005 31	0.005 03	25.481	56
57	93 583	554	0.005 92	0.005 62	24.614	57
58	93 029	614	0.006 60	0.006 26	23.757	58
59	92 415	683	0.007 39	0.007 00	22.912	59
60	91 732	761	0.008 30	0.007 86	22.079	60
61	90 971	839	0.009 22	0.008 80	21.259	61
62	90 132	915	0.010 15	0.009 72	20.452	62
63	89 217	1007	0.011 29	0.010 74	19.657	63
64	88 210	1117	0.012 66	0.012 03	18.875	64
65	87 093	1218	0.013 99	0.013 42	18.111	65
66	85 875	1308	0.015 23	0.014 70	17.361	66
67	84 567	1417	0.016 76	0.016 09	16.621	67
68	83 150	1533	0.018 44	0.017 74	15.896	68
69	81 617	1647	0.020 17	0.019 49	15.185	69
70	79 970	1751	0.021 90	0.021 23	14.487	70
71	78 219	1876	0.023 99	0.023 11	13.800	71
72	76 343	2056	0.026 93	0.025 69	13.127	72
73	74 287	2239	0.030 14	0.028 97	12.476	73
74	72 048	2366	0.032 84	0.032 03	11.848	74
75	69 682	2487	0.035 69	0.034 80	11.234	75
76	67 195	2634	0.039 19	0.038 03	10.631	76
77	64 561	2812	0.043 56	0.042 14	10.044	77
78	61 749	2984	0.048 33	0.046 94	9.478	78
79	58 765	3158	0.053 73	0.052 28	8.934	79
80	55 607	3314	0.059 61	0.058 27	8.413	80
81	52 293	3435	0.065 68	0.064 64	7.914	81
82	48 858	3526	0.072 16	0.071 31	7.435	82
83	45 332	3596	0.079 33	0.078 61	6.974	83
84	41 736	3655	0.087 57	0.086 91	6.532	84
85	38 081	3706	0.097 31	0.096 74	6.111	85
86	34 375	3724	0.108 33	0.108 41	5.715	86
87	30 651	3634	0.118 59	0.120 52	5.349	87
88	27 017	3475	0.128 60	0.131 74	5.002	88
89	23 542	3330	0.141 46	0.144 62	4.667	89
90	20 212	3143	0.155 50	0.160 53	4.354	90
91	17 069	2903	0.170 06	0.177 51	4.065	91
92	14 166	2631	0.185 73	0.195 73	3.797	92
93	11 535	2321	0.201 26	0.214 98	3.551	93
94	9 214	2008	0.217 90	0.234 90	3.322	94
95	7 206	1702	0.236 19	0.257 32	3.112	95
96	5 504	1395	0.253 44	0.281 14	2.925	96
97	4 109	1102	0.268 20	0.302 67	2.754	97
98	3 007	853	0.283 52	0.322 41	2.588	98
99	2 154	653	0.303 31	0.346 28	2.422	99
100	1 501	488	0.324 89	0.376 71	2.269	100
101	1 013	350	0.345 62	0.408 87	2.133	101
102	663	240	0.361 86	0.437 69	2.011	102
103	423	161	0.379 92	0.462 73	1.887	103
104	262	105	0.400 45	0.493 00	1.758	104
105	157	68	0.436 18	0.537 29	1.621	105
106	89	41	0.459 94	0.599 08	1.518	106
107	48	23	0.483 89	0.637 85	1.425	107
108	25	13	0.507 91	0.683 88	1.338	108
109	12	6	0.531 90	0.731 91	1.257	109
110	6	3	0.555 74	0.781 81	1.183	110
111	3	2	0.579 32	0.833 37	1.114	111
112	1	1	0.602 55	0.886 29	1.050	112

AM92

x	$l_{[x]}$	$l_{[x-1]+1}$	l_x	x
17	9 997.809 1		10 000.000 0	17
18	9 991.890 4	9 993.540 0	9 994.000 0	18
19	9 986.035 1	9 987.633 8	9 988.063 6	19
20	9 980.243 2	9 981.791 1	9 982.200 6	20
21	9 974.504 6	9 976.001 6	9 976.390 9	21
22	9 968.839 1	9 970.265 4	9 970.634 6	22
23	9 963.196 7	9 964.582 4	9 964.931 3	23
24	9 957.577 5	9 958.922 5	9 959.261 3	24
25	9 951.991 3	9 953.285 8	9 953.614 4	25
26	9 946.398 2	9 947.662 2	9 947.980 7	26
27	9 940.798 4	9 942.021 8	9 942.340 2	27
28	9 935.181 8	9 936.354 9	9 936.673 0	28
29	9 929.508 8	9 930.661 3	9 930.969 4	29
30	9 923.749 7	9 924.891 6	9 925.209 4	30
31	9 917.914 5	9 919.026 0	9 919.353 5	31
32	9 911.953 8	9 913.054 7	9 913.382 1	32
33	9 905.828 2	9 906.928 5	9 907.265 5	33
34	9 899.498 4	9 900.607 8	9 900.964 5	34
35	9 892.915 1	9 894.053 6	9 894.429 9	35
36	9 886.039 5	9 887.206 9	9 887.612 6	36
37	9 878.812 8	9 880.028 8	9 880.454 0	37
38	9 871.166 5	9 872.450 8	9 872.895 4	38
39	9 863.022 7	9 864.404 7	9 864.868 8	39
40	9 854.303 6	9 855.793 1	9 856.286 3	40
41	9 844.902 5	9 846.538 4	9 847.051 0	41
42	9 834.703 0	9 836.524 5	9 837.066 1	42
43	9 823.599 4	9 825.635 4	9 826.206 0	43
44	9 811.447 3	9 813.746 3	9 814.335 9	44
45	9 798.083 7	9 800.693 9	9 801.312 3	45
46	9 783.337 1	9 786.316 2	9 786.953 4	46
47	9 766.998 3	9 770.423 1	9 771.078 9	47
48	9 748.860 3	9 752.787 4	9 753.471 4	48
49	9 728.649 9	9 733.193 8	9 733.886 5	49
50	9 706.097 7	9 711.352 4	9 712.072 8	50
51	9 680.899 0	9 686.966 9	9 687.714 9	51
52	9 652.696 5	9 659.707 5	9 660.502 1	52
53	9 621.100 6	9 629.211 5	9 630.052 2	53
54	9 585.691 6	9 595.056 3	9 595.971 5	54
55	9 545.992 9	9 556.800 3	9 557.817 9	55
56	9 501.483 9	9 513.937 5	9 515.104 0	56
57	9 451.593 8	9 465.929 3	9 467.290 6	57
58	9 395.697 1	9 412.171 2	9 413.800 4	58
59	9 333.128 4	9 352.016 5	9 354.004 0	59
60	9 263.142 2	9 284.764 1	9 287.216 4	60
61	9 184.968 7	9 209.656 8	9 212.714 3	61
62	9 097.740 5	9 125.881 8	9 129.717 0	62
63	9 000.588 4	9 032.564 2	9 037.397 3	63
64	8 892.574 1	8 928.817 7	8 934.877 1	64

AM92

x	$l_{[x]}$	$l_{[x-1]+1}$	l_x	x
65	8 772.735 9	8 813.688 1	8 821.261 2	65
66	8 640.048 1	8 686.201 6	8 695.619 9	66
67	8 493.518 7	8 545.353 2	8 557.011 8	67
68	8 332.139 6	8 390.161 1	8 404.491 6	68
69	8 154.931 8	8 219.639 0	8 237.132 9	69
70	7 960.977 6	8 032.860 6	8 054.054 4	70
71	7 749.465 9	7 828.968 6	7 854.450 8	71
72	7 519.702 7	7 607.240 0	7 637.620 8	72
73	7 271.146 1	7 367.082 8	7 403.008 4	73
74	7 003.521 6	7 108.105 2	7 150.240 1	74
75	6 716.823 1	6 830.184 4	6 879.167 3	75
76	6 411.345 9	6 533.500 8	6 589.925 8	76
77	6 087.808 4	6 218.575 9	6 282.980 3	77
78	5 747.362 4	5 886.362 8	5 959.168 0	78
79	5 391.640 0	5 538.279 1	5 619.757 7	79
80	5 022.793 1	5 176.222 4	5 266.460 4	80
81	4 643.512 9	4 802.629 0	4 901.478 9	81
82	4 257.005 6	4 420.452 5	4 527.496 0	82
83	3 866.988 4	4 033.146 7	4 147.670 8	83
84	3 477.592 9	3 644.632 7	3 765.599 8	84
85	3 093.286 3	3 259.186 2	3 385.247 9	85
86	2 718.712 8	2 881.346 7	3 010.839 5	86
87	2 358.529 9	2 515.731 0	2 646.741 6	87
88	2 017.229 8	2 166.880 5	2 297.297 6	88
89	1 698.908 9	1 839.045 8	1 966.649 9	89
90	1 407.055 0	1 535.980 1	1 658.554 5	90
91		1 260.735 4	1 376.190 6	91
92			1 121.988 9	92
93			897.502 5	93
94			703.324 2	94
95			539.064 3	95
96			403.402 3	96
97			294.206 1	97
98			208.706 0	98
99			143.712 0	99
100			95.847 6	100
101			61.773 3	101
102			38.379 6	102
103			22.928 4	103
104			13.135 9	104
105			7.196 8	105
106			3.759 6	106
107			1.866 9	107
108			0.878 4	108
109			0.390 3	109
110			0.163 2	110
111			0.064 0	111
112			0.023 4	112
113			0.008 0	113
114			0.002 5	114
115			0.000 7	115
116			0.000 2	116
117			0.000 0	117
118			0.000 0	118
119			0.000 0	119
120			0.000 0	120

AM92

4%

x	$\ddot{a}_{[x]}$	$A_{[x]}$	${}^2A_{[x]}$	$\ddot{a}_x$	A_x	${}^2A_x$	x
65	12.337	0.525 50	0.305 22	12.276	0.527 86	0.308 55	65
66	11.965	0.539 81	0.320 33	11.896	0.542 46	0.324 10	66
67	11.592	0.554 14	0.335 78	11.515	0.557 10	0.340 03	67
68	11.221	0.568 44	0.351 51	11.135	0.571 75	0.356 30	68
69	10.850	0.582 70	0.367 51	10.754	0.586 38	0.372 89	69
70	10.481	0.596 87	0.383 72	10.375	0.600 97	0.389 75	70
71	10.116	0.610 93	0.400 12	9.998	0.615 48	0.406 86	71
72	9.754	0.624 85	0.416 65	9.623	0.629 88	0.424 16	72
73	9.396	0.638 60	0.433 27	9.252	0.644 14	0.441 62	73
74	9.044	0.652 14	0.449 93	8.886	0.658 24	0.459 19	74
75	8.698	0.665 45	0.466 59	8.524	0.672 14	0.476 83	75
76	8.359	0.678 51	0.483 20	8.169	0.685 81	0.494 48	76
77	8.027	0.691 27	0.499 71	7.820	0.699 24	0.512 10	77
78	7.703	0.703 73	0.516 09	7.478	0.712 38	0.529 65	78
79	7.388	0.715 85	0.532 27	7.144	0.725 23	0.547 07	79
80	7.082	0.727 62	0.548 22	6.818	0.737 75	0.564 32	80
81	6.785	0.739 03	0.563 90	6.502	0.749 93	0.581 36	81
82	6.499	0.750 05	0.579 27	6.194	0.761 75	0.598 14	82
83	6.222	0.760 68	0.594 30	5.897	0.773 19	0.614 61	83
84	5.957	0.770 90	0.608 95	5.610	0.784 25	0.630 75	84
85	5.701	0.780 72	0.623 20	5.333	0.794 90	0.646 52	85
86	5.457	0.790 12	0.637 01	5.066	0.805 14	0.661 88	86
87	5.223	0.799 11	0.650 38	4.811	0.814 98	0.676 80	87
88	5.000	0.807 69	0.663 29	4.566	0.824 39	0.691 27	88
89	4.788	0.815 85	0.675 73	4.332	0.833 38	0.705 25	89
90	4.586	0.823 62	0.687 68	4.109	0.841 96	0.718 74	90
91				3.897	0.850 12	0.731 72	91
92				3.695	0.857 87	0.744 17	92
93				3.504	0.865 22	0.756 09	93
94				3.323	0.872 18	0.767 48	94
95				3.153	0.878 75	0.778 34	95
96				2.992	0.884 94	0.788 67	96
97				2.840	0.890 77	0.798 47	97
98				2.698	0.896 25	0.807 76	98
99				2.564	0.901 39	0.816 54	99
100				2.439	0.906 21	0.824 83	100
101				2.321	0.910 71	0.832 63	101
102				2.212	0.914 92	0.839 97	102
103				2.110	0.918 85	0.846 86	103
104				2.015	0.922 51	0.853 31	104
105				1.926	0.925 91	0.859 34	105
106				1.844	0.929 07	0.864 98	106
107				1.768	0.932 01	0.870 23	107
108				1.697	0.934 72	0.875 12	108
109				1.632	0.937 24	0.879 66	109
110				1.571	0.939 56	0.883 87	110
111				1.516	0.941 70	0.887 77	111
112				1.464	0.943 67	0.891 37	112
113				1.417	0.945 49	0.894 69	113
114				1.374	0.947 15	0.897 75	114
115				1.334	0.948 68	0.900 56	115
116				1.298	0.950 08	0.903 15	116
117				1.264	0.951 39	0.905 57	117
118				1.229	0.952 73	0.908 04	118
119				1.176	0.954 78	0.911 81	119
120				1.000	0.961 54	0.924 56	120

Note. ${}^2A_{[x]} = A_{[x]}$ at 8.16% and ${}^2A_x = A_x$ at 8.16%.

AM92

4%

x	$\ddot{a}_{[x]:\overline{n}\rceil}$	$A_{[x]:\overline{n}\rceil}$	$n = 60 - x$	$\ddot{a}_{x:\overline{n}\rceil}$	$A_{x:\overline{n}\rceil}$	x
17	20.941	0.194 59	43	20.936	0.194 75	17
18	20.750	0.201 90	42	20.746	0.202 06	18
19	20.552	0.209 53	41	20.548	0.209 68	19
20	20.346	0.217 46	40	20.342	0.217 60	20
21	20.131	0.225 72	39	20.128	0.225 86	21
22	19.908	0.234 32	38	19.904	0.234 45	22
23	19.675	0.243 27	37	19.672	0.243 40	23
24	19.433	0.252 59	36	19.430	0.252 71	24
25	19.181	0.262 28	35	19.178	0.262 40	25
26	18.918	0.272 37	34	18.916	0.272 48	26
27	18.645	0.282 87	33	18.643	0.282 97	27
28	18.361	0.293 79	32	18.359	0.293 89	28
29	18.066	0.305 15	31	18.064	0.305 25	29
30	17.759	0.316 97	30	17.756	0.317 06	30
31	17.439	0.329 26	29	17.437	0.329 35	31
32	17.107	0.342 04	28	17.105	0.342 12	32
33	16.762	0.355 33	27	16.759	0.355 41	33
34	16.402	0.369 14	26	16.400	0.369 23	34
35	16.029	0.383 50	25	16.027	0.383 59	35
36	15.641	0.398 43	24	15.639	0.398 52	36
37	15.237	0.413 95	23	15.235	0.414 03	37
38	14.818	0.430 07	22	14.816	0.430 16	38
39	14.383	0.446 82	21	14.380	0.446 92	39
40	13.930	0.464 23	20	13.927	0.464 33	40
41	13.460	0.482 31	19	13.457	0.482 42	41
42	12.971	0.501 10	18	12.969	0.501 21	42
43	12.464	0.520 61	17	12.461	0.520 73	43
44	11.937	0.540 88	16	11.934	0.541 00	44
45	11.390	0.561 93	15	11.386	0.562 06	45
46	10.821	0.583 80	14	10.818	0.583 93	46
47	10.231	0.606 51	13	10.227	0.606 65	47
48	9.617	0.630 10	12	9.613	0.630 25	48
49	8.980	0.654 61	11	8.976	0.654 77	49
50	8.318	0.680 07	10	8.314	0.680 24	50
51	7.630	0.706 54	9	7.625	0.706 72	51
52	6.914	0.734 06	8	6.910	0.734 24	52
53	6.170	0.762 68	7	6.166	0.762 86	53
54	5.396	0.792 46	6	5.391	0.792 64	54
55	4.590	0.823 48	5	4.585	0.823 65	55
56	3.749	0.855 80	4	3.745	0.855 95	56
57	2.873	0.889 52	3	2.870	0.889 63	57
58	1.957	0.924 73	2	1.955	0.924 79	58
59	1.000	0.961 54	1	1.000	0.961 54	59

AM92

4%

x	$\ddot{a}_{[x]:\overline{n}\rceil}$	$A_{[x]:\overline{n}\rceil}$	$n = 65 - x$	$\ddot{a}_{x:\overline{n}\rceil}$	$A_{x:\overline{n}\rceil}$	x
17	21.723	0.164 48	48	21.719	0.164 66	17
18	21.565	0.170 58	47	21.561	0.170 74	18
19	21.400	0.176 93	46	21.396	0.177 09	19
20	21.228	0.183 54	45	21.224	0.183 69	20
21	21.049	0.190 42	44	21.045	0.190 57	21
22	20.863	0.197 59	43	20.859	0.197 73	22
23	20.669	0.205 05	42	20.665	0.205 18	23
24	20.467	0.212 81	41	20.464	0.212 94	24
25	20.257	0.220 90	40	20.254	0.221 02	25
26	20.038	0.229 31	39	20.035	0.229 42	26
27	19.811	0.238 05	38	19.808	0.238 17	27
28	19.574	0.247 16	37	19.571	0.247 26	28
29	19.328	0.256 62	36	19.325	0.256 73	29
30	19.072	0.266 47	35	19.069	0.266 57	30
31	18.806	0.276 71	34	18.803	0.276 81	31
32	18.529	0.287 35	33	18.526	0.287 45	32
33	18.241	0.298 42	32	18.239	0.298 52	33
34	17.942	0.309 92	31	17.940	0.310 02	34
35	17.631	0.321 87	30	17.629	0.321 97	35
36	17.308	0.334 29	29	17.306	0.334 39	36
37	16.973	0.347 19	28	16.970	0.347 29	37
38	16.625	0.360 59	27	16.622	0.360 70	38
39	16.263	0.374 51	26	16.260	0.374 62	39
40	15.887	0.388 96	25	15.884	0.389 07	40
41	15.497	0.403 95	24	15.494	0.404 07	41
42	15.092	0.419 52	23	15.089	0.419 65	42
43	14.672	0.435 67	22	14.669	0.435 81	43
44	14.237	0.452 43	21	14.233	0.452 58	44
45	13.785	0.469 82	20	13.780	0.469 98	45
46	13.316	0.487 86	19	13.311	0.488 03	46
47	12.829	0.506 56	18	12.824	0.506 75	47
48	12.325	0.525 96	17	12.320	0.526 17	48
49	11.802	0.546 08	16	11.796	0.546 30	49
50	11.259	0.566 95	15	11.253	0.567 19	50
51	10.697	0.588 58	14	10.690	0.588 84	51
52	10.113	0.611 02	13	10.106	0.611 30	52
53	9.508	0.634 30	12	9.500	0.634 60	53
54	8.880	0.658 46	11	8.872	0.658 78	54
55	8.228	0.683 54	10	8.219	0.683 88	55
56	7.551	0.709 58	9	7.542	0.709 93	56
57	6.847	0.736 64	8	6.838	0.737 01	57
58	6.115	0.764 79	7	6.106	0.765 16	58
59	5.353	0.794 10	6	5.344	0.794 46	59
60	4.559	0.824 65	5	4.550	0.824 99	60
61	3.730	0.856 54	4	3.722	0.856 85	61
62	2.863	0.889 90	3	2.857	0.890 13	62
63	1.954	0.924 85	2	1.951	0.924 98	63
64	1.000	0.961 54	1	1.000	0.961 54	64

AM92

6%

x	$\ddot{a}_{[x]}$	$A_{[x]}$	${}^2A_{[x]}$	$\ddot{a}_x$	A_x	${}^2A_x$	x
17	16.977	0.039 02	0.006 11	16.974	0.039 21	0.006 30	17
18	16.946	0.040 80	0.006 30	16.943	0.040 99	0.006 48	18
19	16.912	0.042 70	0.006 52	16.909	0.042 88	0.006 69	19
20	16.877	0.044 72	0.006 77	16.874	0.044 89	0.006 93	20
21	16.839	0.046 86	0.007 05	16.836	0.047 03	0.007 21	21
22	16.798	0.049 14	0.007 38	16.796	0.049 30	0.007 53	22
23	16.756	0.051 57	0.007 75	16.753	0.051 72	0.007 90	23
24	16.710	0.054 14	0.008 16	16.708	0.054 28	0.008 31	24
25	16.662	0.056 86	0.008 63	16.660	0.057 01	0.008 77	25
26	16.611	0.059 76	0.009 16	16.609	0.059 90	0.009 30	26
27	16.557	0.062 82	0.009 75	16.554	0.062 96	0.009 88	27
28	16.499	0.066 07	0.010 41	16.497	0.066 20	0.010 54	28
29	16.439	0.069 51	0.011 15	16.436	0.069 64	0.011 28	29
30	16.374	0.073 16	0.011 97	16.372	0.073 28	0.012 10	30
31	16.306	0.077 01	0.012 89	16.304	0.077 14	0.013 01	31
32	16.234	0.081 09	0.013 90	16.232	0.081 21	0.014 03	32
33	16.158	0.085 40	0.015 03	16.156	0.085 52	0.015 15	33
34	16.078	0.089 95	0.016 27	16.075	0.090 07	0.016 40	34
35	15.993	0.094 75	0.017 65	15.990	0.094 88	0.017 78	35
36	15.903	0.099 82	0.019 16	15.901	0.099 95	0.019 30	36
37	15.809	0.105 16	0.020 84	15.806	0.105 30	0.020 98	37
38	15.709	0.110 79	0.022 67	15.707	0.110 94	0.022 82	38
39	15.605	0.116 72	0.024 69	15.602	0.116 88	0.024 85	39
40	15.494	0.122 96	0.026 90	15.491	0.123 13	0.027 07	40
41	15.378	0.129 52	0.029 33	15.375	0.129 70	0.029 51	41
42	15.257	0.136 41	0.031 98	15.253	0.136 60	0.032 18	42
43	15.129	0.143 65	0.034 87	15.125	0.143 85	0.035 09	43
44	14.995	0.151 23	0.038 02	14.991	0.151 46	0.038 26	44
45	14.855	0.159 18	0.041 45	14.850	0.159 43	0.041 72	45
46	14.708	0.167 50	0.045 17	14.703	0.167 78	0.045 48	46
47	14.554	0.176 19	0.049 21	14.548	0.176 51	0.049 56	47
48	14.393	0.185 28	0.053 59	14.387	0.185 63	0.053 98	48
49	14.226	0.194 76	0.058 32	14.219	0.195 16	0.058 76	49
50	14.051	0.204 63	0.063 42	14.044	0.205 08	0.063 92	50
51	13.870	0.214 91	0.068 92	13.861	0.215 42	0.069 49	51
52	13.681	0.225 60	0.074 83	13.671	0.226 17	0.075 48	52
53	13.485	0.236 69	0.081 18	13.474	0.237 34	0.081 92	53
54	13.282	0.248 18	0.087 97	13.269	0.248 92	0.088 82	54
55	13.072	0.260 08	0.095 24	13.057	0.260 92	0.096 21	55
56	12.855	0.272 37	0.102 98	12.838	0.273 33	0.104 09	56
57	12.631	0.285 06	0.111 23	12.612	0.286 14	0.112 50	57
58	12.400	0.298 12	0.119 98	12.378	0.299 35	0.121 44	58
59	12.163	0.311 55	0.129 26	12.138	0.312 94	0.130 93	59
60	11.919	0.325 33	0.139 07	11.891	0.326 92	0.140 98	60
61	11.670	0.339 45	0.149 41	11.638	0.341 25	0.151 60	61
62	11.415	0.353 88	0.160 29	11.379	0.355 92	0.162 80	62
63	11.155	0.368 61	0.171 71	11.114	0.370 91	0.174 57	63
64	10.890	0.383 60	0.183 66	10.844	0.386 20	0.186 92	64

Note. ${}^2A_{[x]} = A_{[x]}$ at 12.36% and ${}^2A_x = A_x$ at 12.36%.

AM92

6%

x	$\ddot{a}_{[x]}$	$A_{[x]}$	${}^2A_{[x]}$	$\ddot{a}_x$	A_x	${}^2A_x$	x
65	10.621	0.398 83	0.196 14	10.569	0.401 77	0.199 85	65
66	10.348	0.414 27	0.209 13	10.289	0.417 58	0.213 35	66
67	10.072	0.429 88	0.222 62	10.006	0.433 61	0.227 40	67
68	9.794	0.445 64	0.236 58	9.720	0.449 82	0.242 00	68
69	9.513	0.461 50	0.251 00	9.431	0.466 17	0.257 12	69
70	9.232	0.477 43	0.265 83	9.140	0.482 65	0.272 74	70
71	8.950	0.493 38	0.281 06	8.848	0.499 19	0.288 82	71
72	8.669	0.509 33	0.296 64	8.555	0.515 78	0.305 34	72
73	8.388	0.525 21	0.312 54	8.262	0.532 36	0.322 26	73
74	8.109	0.541 01	0.328 70	7.969	0.548 90	0.339 55	74
75	7.832	0.556 67	0.345 09	7.679	0.565 35	0.357 14	75
76	7.559	0.572 15	0.361 64	7.390	0.581 69	0.375 01	76
77	7.289	0.587 42	0.378 33	7.105	0.597 86	0.393 09	77
78	7.024	0.602 44	0.395 08	6.822	0.613 83	0.411 33	78
79	6.763	0.617 17	0.411 86	6.544	0.629 56	0.429 69	79
80	6.509	0.631 59	0.428 60	6.271	0.645 01	0.448 11	80
81	6.260	0.645 66	0.445 25	6.004	0.660 16	0.466 52	81
82	6.018	0.659 35	0.461 77	5.742	0.674 97	0.484 88	82
83	5.783	0.672 65	0.478 11	5.487	0.689 42	0.503 13	83
84	5.556	0.685 53	0.494 22	5.239	0.703 46	0.521 21	84
85	5.336	0.697 97	0.510 05	4.998	0.717 10	0.539 07	85
86	5.124	0.709 97	0.525 57	4.765	0.730 29	0.556 67	86
87	4.920	0.721 50	0.540 75	4.540	0.743 04	0.573 96	87
88	4.724	0.732 58	0.555 55	4.323	0.755 31	0.590 88	88
89	4.537	0.743 18	0.569 94	4.114	0.767 11	0.607 41	89
90	4.358	0.753 32	0.583 90	3.914	0.778 43	0.623 50	90
91				3.723	0.789 25	0.639 13	91
92				3.541	0.799 59	0.654 26	92
93				3.367	0.809 44	0.668 88	93
94				3.201	0.818 80	0.682 96	94
95				3.044	0.827 69	0.696 49	95
96				2.896	0.836 10	0.709 46	96
97				2.755	0.844 06	0.721 87	97
98				2.622	0.851 56	0.733 70	98
99				2.498	0.858 63	0.744 96	99
100				2.380	0.865 27	0.755 65	100
101				2.270	0.871 51	0.765 79	101
102				2.167	0.877 36	0.775 37	102
103				2.070	0.882 83	0.784 42	103
104				1.980	0.887 94	0.792 93	104
105				1.895	0.892 71	0.800 94	105
106				1.817	0.897 15	0.808 45	106
107				1.744	0.901 28	0.815 48	107
108				1.676	0.905 11	0.822 05	108
109				1.614	0.908 66	0.828 17	109
110				1.556	0.911 95	0.833 87	110
111				1.502	0.914 99	0.839 17	111
112				1.452	0.917 79	0.844 08	112
113				1.407	0.920 37	0.848 61	113
114				1.365	0.922 75	0.852 80	114
115				1.326	0.924 92	0.856 66	115
116				1.291	0.926 93	0.860 22	116
117				1.258	0.928 80	0.863 55	117
118				1.224	0.930 72	0.866 94	118
119				1.172	0.933 64	0.872 10	119
120				1.000	0.943 40	0.890 00	120

Note. ${}^2A_{[x]} = A_{[x]}$ at 12.36% and ${}^2A_x = A_x$ at 12.36%.

AM92

6%

x	$\ddot{a}_{[x]:\overline{n}\rceil}$	$A_{[x]:\overline{n}\rceil}$	$n = 60 - x$	$\ddot{a}_{x:\overline{n}\rceil}$	$A_{x:\overline{n}\rceil}$	x
17	16.076	0.090 05	43	16.072	0.090 24	17
18	15.990	0.094 93	42	15.986	0.095 11	18
19	15.898	0.100 11	41	15.895	0.100 28	19
20	15.801	0.105 61	40	15.798	0.105 77	20
21	15.698	0.111 45	39	15.695	0.111 60	21
22	15.588	0.117 64	38	15.586	0.117 79	22
23	15.472	0.124 22	37	15.470	0.124 36	23
24	15.349	0.131 19	36	15.347	0.131 33	24
25	15.218	0.138 59	35	15.216	0.138 72	25
26	15.080	0.146 43	34	15.078	0.146 56	26
27	14.933	0.154 75	33	14.931	0.154 87	27
28	14.777	0.163 57	32	14.775	0.163 69	28
29	14.612	0.172 92	31	14.610	0.173 03	29
30	14.437	0.182 83	30	14.435	0.182 94	30
31	14.251	0.193 33	29	14.249	0.193 44	31
32	14.054	0.204 46	28	14.053	0.204 57	32
33	13.846	0.216 26	27	13.844	0.216 36	33
34	13.625	0.228 75	26	13.624	0.228 85	34
35	13.392	0.241 98	25	13.390	0.242 08	35
36	13.144	0.255 99	24	13.142	0.256 09	36
37	12.882	0.270 82	23	12.880	0.270 93	37
38	12.605	0.286 53	22	12.603	0.286 64	38
39	12.311	0.303 16	21	12.309	0.303 27	39
40	12.000	0.320 76	20	11.998	0.320 88	40
41	11.671	0.339 38	19	11.669	0.339 51	41
42	11.323	0.359 10	18	11.320	0.359 23	42
43	10.954	0.379 96	17	10.952	0.380 10	43
44	10.564	0.402 03	16	10.561	0.402 19	44
45	10.151	0.425 39	15	10.149	0.425 56	45
46	9.715	0.450 11	14	9.712	0.450 28	46
47	9.253	0.476 26	13	9.249	0.476 45	47
48	8.764	0.503 94	12	8.760	0.504 15	48
49	8.246	0.533 24	11	8.242	0.533 46	49
50	7.698	0.564 26	10	7.694	0.564 49	50
51	7.118	0.597 11	9	7.114	0.597 35	51
52	6.503	0.631 91	8	6.499	0.632 16	52
53	5.851	0.668 79	7	5.847	0.669 04	53
54	5.160	0.707 91	6	5.156	0.708 15	54
55	4.427	0.749 41	5	4.423	0.749 65	55
56	3.648	0.793 50	4	3.645	0.793 70	56
57	2.820	0.840 36	3	2.817	0.840 52	57
58	1.939	0.890 24	2	1.937	0.890 34	58
59	1.000	0.943 40	1	1.000	0.943 40	59

AM92

6%

x	$\ddot{a}_{[x]:\overline{n}\rceil}$	$A_{[x]:\overline{n}\rceil}$	$n = 65 - x$	$\ddot{a}_{x:\overline{n}\rceil}$	$A_{x:\overline{n}\rceil}$	x
17	16.409	0.071 21	48	16.405	0.071 40	17
18	16.343	0.074 95	47	16.339	0.075 13	18
19	16.272	0.078 92	46	16.269	0.079 09	19
20	16.198	0.083 13	45	16.195	0.083 30	20
21	16.119	0.087 61	44	16.116	0.087 77	21
22	16.035	0.092 36	43	16.032	0.092 51	22
23	15.946	0.097 40	42	15.943	0.097 54	23
24	15.852	0.102 74	41	15.849	0.102 88	24
25	15.751	0.108 42	40	15.749	0.108 55	25
26	15.645	0.114 43	39	15.643	0.114 56	26
27	15.532	0.120 81	38	15.530	0.120 94	27
28	15.413	0.127 58	37	15.411	0.127 70	28
29	15.286	0.134 75	36	15.284	0.134 86	29
30	15.152	0.142 34	35	15.150	0.142 46	30
31	15.010	0.150 39	34	15.008	0.150 50	31
32	14.859	0.158 92	33	14.857	0.159 03	32
33	14.700	0.167 95	32	14.698	0.168 06	33
34	14.531	0.177 51	31	14.529	0.177 62	34
35	14.352	0.187 63	30	14.350	0.187 74	35
36	14.163	0.198 33	29	14.161	0.198 45	36
37	13.963	0.209 67	28	13.960	0.209 79	37
38	13.751	0.221 65	27	13.749	0.221 78	38
39	13.527	0.234 33	26	13.525	0.234 46	39
40	13.290	0.247 74	25	13.288	0.247 87	40
41	13.040	0.261 91	24	13.037	0.262 06	41
42	12.775	0.276 89	23	12.772	0.277 05	42
43	12.495	0.292 72	22	12.492	0.292 89	43
44	12.200	0.309 44	21	12.197	0.309 63	44
45	11.888	0.327 11	20	11.884	0.327 31	45
46	11.558	0.345 78	19	11.554	0.345 99	46
47	11.210	0.365 49	18	11.206	0.365 72	47
48	10.842	0.386 30	17	10.837	0.386 56	48
49	10.454	0.408 28	16	10.449	0.408 57	49
50	10.044	0.431 50	15	10.038	0.431 81	50
51	9.610	0.456 02	14	9.604	0.456 35	51
52	9.153	0.481 91	13	9.146	0.482 28	52
53	8.669	0.509 27	12	8.662	0.509 67	53
54	8.159	0.538 19	11	8.151	0.538 62	54
55	7.618	0.568 77	10	7.610	0.569 22	55
56	7.047	0.601 12	9	7.038	0.601 60	56
57	6.442	0.635 36	8	6.433	0.635 86	57
58	5.801	0.671 65	7	5.792	0.672 16	58
59	5.121	0.710 15	6	5.112	0.710 66	59
60	4.398	0.751 04	5	4.390	0.751 52	60
61	3.630	0.794 54	4	3.622	0.794 97	61
62	2.811	0.840 90	3	2.805	0.841 23	62
63	1.936	0.890 42	2	1.933	0.890 60	63
64	1.000	0.943 40	1	1.000	0.943 40	64